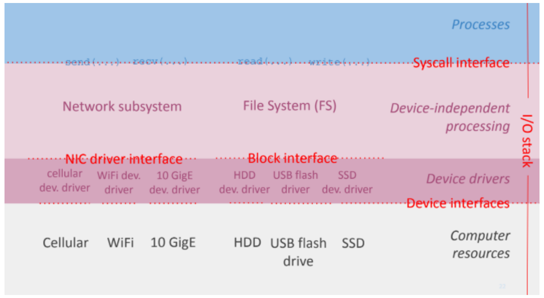
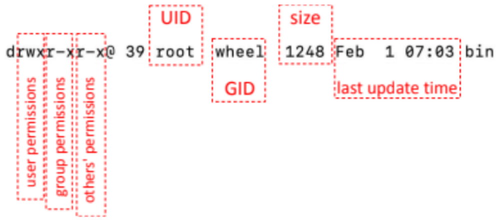
Computer Systems · BA4

Protocols / Technologies	Layer
HTTP, DNS, BitTorrent	Application
TCP, UDP	Transport
IP	Network
WiFi, Ethernet, FTTH	Link
EM waves, optical, electric	Physical



1 Lecture 5 - Network & File Syscalls



Permission bits

Am I the **owner**? → use USR bits.
Am I in the **group**? → use GRP bits.
Otherwise → use OTH bits.
Only the first matching category applies — even if another gives more rights.

open() flags / permission mode

`O_CREAT`	create file if it doesn't exist
`O_EXCL`	fail if file already exists
`O_APPEND`	always write at end of file
`O_TRUNC`	truncate file on open
`O_RDONLY`	read only
`O_WRONLY`	write only
`O_RDWR`	read + write
`S_IRUSR`	read by owner
`S_IWUSR`	write by owner
`S_IXUSR`	execute by owner
`S_IRGRP`	read by group
`S_IROTH`	read by others
`0`	no permissions

Syscall success conditions

`open(O_RDONLY)`	`r` on file
`open(O_WRONLY)`	`w` on file
`open(O_RDWR)`	`r+w` on file
`open(O_CREAT)`	`w+x` on dir
`read(fd,...)`	fd open, mode=read
`write(fd,...)`	fd open, mode=write
`close(fd)`	fd open

Also: **every directory** in the path needs `x` to be traversed.

Closed fd

Any syscall on a **closed** fd fails immediately.

S_* flags

Only used with `O_CREAT` to set the new file's permissions. **Ignored** otherwise.

2 Lecture 8 - Forwarding and IP

Internet (IP) forwarding

dest. address range			output link
0 - 2	0000 - 0010	00**	1
3	0011	0011	2
4, 6, 7	0100, 0110, 0111	01**	3
5	0101	0101	2
8 - 15	1000 - 1111	1***	4

Each entry is a **prefix** (e.g. `01**`); a packet is forwarded using the **longest matching prefix**.

IP prefix

`a.b.c.d/n` denotes a range of addresses sharing the same first n bits (prefix length); the remaining $32-n$ bits are wildcards.

Broadcast IP address (+1)

The `biggest IP address` an IP subnet. If a packet has this destination, it's destined to **all the end-systems** and **routers** in the IP subnet.

Internet principle

Global reachability: Every end-system must be reachable from any other end-system.

Network Address Translation (NAT)

A router becomes a NAT only when it's configured to translate between private and public addresses.

Layering violation (but it works)

- Requires **per-connection state** at the border routers of the private IP subnets.
- An end-system inside a private IP subnet is **unreachable** from the outside world. [trick possible]

3 Lecture 9 - Routing & BGP

First-hop router

First router packets hit when leaving a subnet.

Internet Routing algorithms

Dijkstra's (Link state routing algo)

1. At each step, consider a new router starting from "closest" neighbor
2. Check whether current paths can be improved
3. End when no improvement is possible

Bellman-Ford (Distance vector)

1. All neighbors exchange information
2. Each router checks whether it can improve current paths by leveraging the new information
3. End when no improvement is possible

```
INIT-SINGLE-SOURCE(G, s)
    s.d = 0
    for v in G.V: v.d = inf; v.pi = NIL

RELAX(u, v, w)
    if v.d > u.d + w(u, v)
        v.d = u.d + w(u, v); v.pi = u
```

```
BELLMAN-FORD(G, w, s)
    INIT-SINGLE-SOURCE(G, s)
    for i = 1 to |G.V| - 1
        for (u, v) in G.E:
            RELAX(u, v, w)
```

```
DIJKSTRA(G, w, s)
    INIT-SINGLE-SOURCE(G, s)
    S = {}
    Q = G.V   // min-priority queue (sorted by d(v))
    while Q != {}:
        u = EXTRACT-MIN(Q)
        S = S U {u}
        for each vertex v in G.Adj[u]:
            RELAX(u, v, w)
```

Link state VS Distance vector

- Link state converges faster
- Distance vector uses less bandwitdh

Internet rounting challenges

- **Admin:** An ISP may want to hide its link costs
- **Scale:** Too many IP subnets may cause trouble

Autonomous System (AS)

A network or group of networks under a **single administrative authority**, running its own internal routing protocol. Identified by a unique **ASN**.

Inter-AS Routing (BGP)

Routing **between** ASes. One universal protocol based on Bellman-Ford. Not always the shortest path.

Border Gateway Protocol (BGP)

1. All neighbors exchange reachable prefixes
2. Each router checks whether it can improve paths
3. End when no improvement is possible

Internet rounting solutions

- Router ↔ with **local** router
- Border router ↔ with **external neighbor**
- Router keeps **forwarding state** for (1) **local IP subnets** and (2) **foreign aggregate IP prefixe**

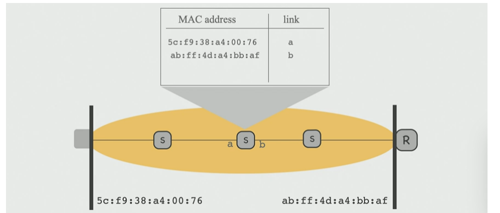
4 Lecture 10 - Link layer & Ethernet

Link layer services

- Error detection & Reliable data delivery
- *Medium access control (MAC)*

Link layer header: MAC address

48-bit. Flat: not location dependent



Spanning tree Broadcast

The naïve broadcast may lead to forwarding loops. Use spanning tree to avoid cycles.

Address Resolution Protocol (ARP)

Find the MAC address associated to an IP (to build the L2 frame) via broadcast + caching.

5 Lecture 11 - File Systems

Persistence

State that **outlives the process that created it** (unlike DRAM, which is volatile).

File System

Abstracts persistent blocks into: **Name**, **Organize**, **Share**, **Protect**. Built on **files** and **directories**.

File

Named array of bytes, untyped. Has **data** + **metadata** (size, owner, permissions, timestamps).

Inode

Fixed-size on-disk record: **metadata + pointers to data blocks**. One per file. Stored in a fixed array (inode table, indexed by number). Does **not** store the filename.

Multi-level indexing

$N = B/P$ (ptrs/block). Max size $= kB + NB + N^2B + N^3B$ (direct + 1/2/3-indirect).

Directory

Special file containing an array of **{filename, inode}** pairs. Path resolution follows entries step-by-step.

File Descriptor Table

Per-process table caching the result of `open()`. Fixed entries: **0**=STDIN, **1**=STDOUT, **2**=STDERR. Each entry stores **inode**, **mode**, **offset**. Each `open()` creates an independent entry with its own offset.

Block Allocation Strategies

Contiguous: all blocks adjacent.
Linked list: each block points to the next.
FAT (File Allocation Table): linked list stored **in a separate table** in memory. Table can be large.

`mv` vs `cp`

Moving a file only updates **directory entries** — the inode is unchanged. Metadata-only ⇒ much faster than `cp + rm`.

Mounting

Integrates a storage device into the directory tree without moving its contents.

Inode & Data Bitmap

`bitmap[i] = 0` → free, `1` → used.

Disk Layout

Boot block, Superblock, Inode Bitmap, Data Bitmap, Inode Table, Data Blocks.

6 Lecture 12 - Processes and Threads

Process

An instance of a running program. Has its own **isolated memory space** (code, data, heap, stack) and OS resources (file descriptors, etc.).

Thread

A unit of execution **within** a process. Threads share the process's memory and resources, but each has its own **stack** and **program counter**.

Key properties

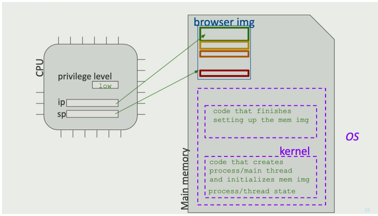
- **Identifier**: unique id for processes and threads
- Each Thread/Process has the **illusion** that is it the only one occupying the CPU/Memory (resp.)

7 Lecture 13 - Sharing the CPU

CPU mode & instructions

CPU stores a privilege level: **Kernel mode** (high) or **User mode** (low).

- **Non-privileged:** executable anytime
- **Privileged:** only in kernel mode



How threads can share the CPU **safely** and **fairly**?

Limited direct execution (invoking the kernel)

Syscall / Exception / Interrupt

How are processes created and destroyed?

Process lifecycle

- `fork`: creates a **child** process (copy of parent)
- `exec`: replaces process image with a new program
- `exit`: terminates the process
- `wait`: parent waits for child to finish

Type of system processes

- **Init**: first process after boot
- **Idle**: runs when nothing else can
- **Shell**: spawns new process on command
- **GUI manager**: spawns new process on click

8 Lecture 14 - Memory & Loose Ends

Memory image

Consists of: **code**, **data**, **heap**, **stack**.

Memory virtualization

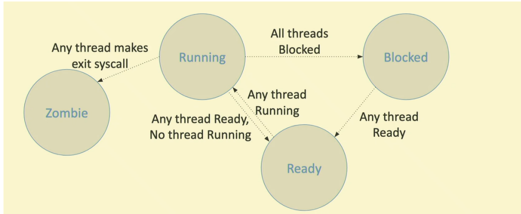
Safe illusion

The process thinks it's occupying the whole memory alone and can only touch its own memory image

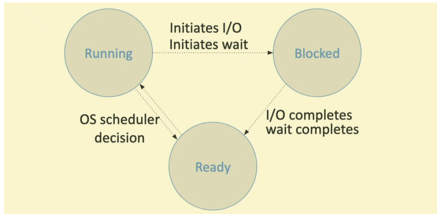
Memory Management Unit (MMU)

- **Base/bound:** base and bound in the memory
- **Segmentation:** variable-sized chunks (segments)
- **Paging:** same-sized chunks (frames)

Process state diagram



Thread state diagram



End of a process

`exit()` syscall: kernel destroys memory image but **keeps exit status** ⇒ process becomes a **zombie** until parent **reaps** it via `wait()` to read the exit status.

9 Lecture 15 - CPU Scheduling

Scheduling Metrics

Turnaround time = completion - arrival.
Response time = first run - arrival.

FIFO

Run jobs in arrival order. Suffers from **convoy effect**: short jobs stuck behind long ones.

SJF (Shortest Job First)

Run shortest job first. Optimal turnaround if all jobs arrive simultaneously. Non-preemptive.

STCF (Shortest Time-to-Completion First)

Preemptive SJF: preempt if a new job has less remaining time. Optimal turnaround.

Round Robin (RR)

Each job runs for a fixed **quantum**, then preempted. Good response time, poor turnaround.

Tradeoff

SJF/STCF optimizes **turnaround**. RR optimizes **response time**. Hard to do both.

MLFQ (Multi-Level Feedback Queue)

Multiple queues; lower priority = longer time slice.

- **R1:** priority(A) > priority(B) ⇒ A runs
- **R2:** priority(A) = priority(B) ⇒ RR
- **R3:** new job starts at **top** priority
- **R4:** uses full slice ⇒ **demoted**; blocks early ⇒ **keeps** priority
- **R5:** periodically move all jobs to top (**priority boost**) — avoids starvation

10 Lecture 16 - Virtualization

Virtualization types

- **Multiplexing:** one resource → multiple virtual entities (e.g. CPU, memory)
- **Aggregation:** multiple resources → one virtual resource (e.g. disk volumes)
- **Emulation:** one physical resource appears as another (e.g. serial port over USB/SSH)

11 Lecture 17/18 - Concurrency

Common confusion

- **Concurrency:** how should we manage tasks
- **Parallism:** how to execute tasks in parallel

Race condition & Data race

- **Race cond.:** behavior depends on thread timing
- **Data race:** unsynchronized concurrent access to shared writable state

Critical section

Code that will be executed by the threads. Must enforce: **atomicity** (uninterruptible) and **mutual exclusion** (one thread at a time).

Synchronization primitives

- **Lock** → prevent concurrent access
- **Condition variable** → wait until a condition holds
- **Semaphore** → count available resources

12 Units

Powers of ten

- 1 KB (kilobyte) $= 10^3 = 1\,000$ bytes
- 1 MB (megabyte) $= 10^6 = 1\,000\,000$ bytes
- 1 GB (gigabyte) $= 10^9 = 1\,000\,000\,000$ bytes

Powers of two

- 1 KiB (kibibyte) $= 2^{10} = 1\,024$ bytes
- 1 MiB (mebibyte) $= 2^{20} = 1\,048\,576$ bytes
- 1 GiB (gibibyte) $= 2^{30} = 1\,073\,741\,824$ bytes